Q-1 Create Database and Insert Records

```
sqlite3 college.db
CREATE TABLE students (id INTEGER PRIMARY KEY,name TEXT NOT NULL,age INTEGER,course
TEXT);
INSERT INTO students VALUES('Rahul', 20, 'BCA'),('Priya', 21, 'BBA'),('Amit', 19,
'BCA');
INSERT INTO students VALUES ('Amit', 19, 'BCA');
SELECT*FROM students;
```

Q-2 Backup Complete Database (.dump equivalent)

```
.dump
.quit
sqlite3 college.db ".dump" > college_backup.sql
```

Q-3 Backup Only Database Structure

```
sqlite3 college.db
.schema
.quit
sqlite3 college.db ".schema" > college_structure.sql
```

Q-4 Backup Only Data

```
sqlite3 college.db
.mode insert
.output college_data.sql
SELECT * FROM students;
.output stdout
.quit
sqlite3 new_college.db < college_data.sql
sqlite3 new_college.db
```

Q-5 Restore Database from Backup

```
sqlite3 restored_college.db < college_backup.sql
sqlite3 restored_college.db
.tables
SELECT*FROM students;
```

Q-6 Export Complete Database to Another File

```
sqlite3 college.db ".backup college_copy.db"
sqlite3 college_copy.db
.tables
SELECT*FROM students;
```

Q-7 Display All Tables

```
sqlite3 college.db
.tables
```

Q-8 Display Database Schema

```
sqlite3 college.db
.schema
.schema students
```

```
Q-9 Count Total Records
sqlite3 college.db
SELECT COUNT(*) FROM students;
SELECT COUNT(*) AS total_records FROM students;


export
sqlite3 college.db
create and insert
CREATE TABLE stud (
    id INTEGER,
    name TEXT,
    age INTEGER,
    course TEXT
);

INSERT INTO stud VALUES (1, 'Jiya', 20, 'BSc CS');
INSERT INTO stud VALUES (2, 'Riya', 21, 'BSc CS');
INSERT INTO stud VALUES (3, 'Neha', 20, 'BCA');

.mode csv
.headers on
.output filename.csv
id,name,age,course
1,Jiya,20,BSc CS
2,Riya,21,BSc CS
3,Neha,20,BCA
select * from stud;
.output stdout

import
kdata.csv
id,name,age,course
4,Krisha,20,BSc CS
5,Hetvi,21,BCA
6,Diya,20,BSc IT
create csv and sqlite3 table same headers on that
CREATE TABLE stud (
    id INTEGER,
    name TEXT,
    age INTEGER,
    course TEXT
);

.mode csv
.import --skip 1 kdata.csv stud
select * from stud;
```

```
mongodb
mongosh
show dbs
use clg
db.createcollection("stud");
db.stud.insertOne
db.stud.insertMany


aggrestion
db.stud.aggregate([{$match:{age:{$gte:18}}}]);
([{$group:{_id:"city",totalstud:{$sum:1}}}]);
([{$group:{_id:"gender",avgage:{$avg:"$age"}}}])

sqlite3
sqlite3 college.db
.database
.table
```